

FALL 2025-2026

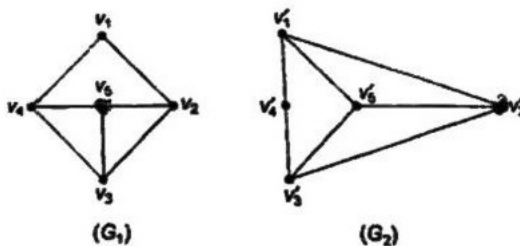
BMAT205L - DISCRETE MATHEMATICS AND GRAPH THEORY

Module-5 and 6 : Tutorial -7

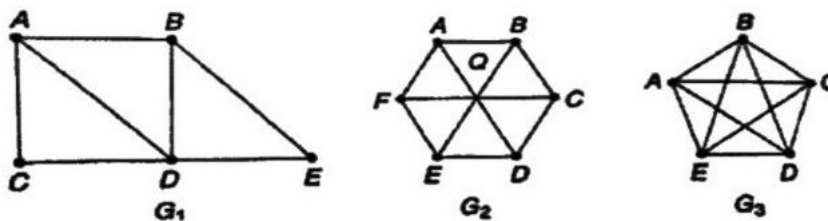
(To assess the students' understanding on Fundamentals of Graphs)

6 November, 2025

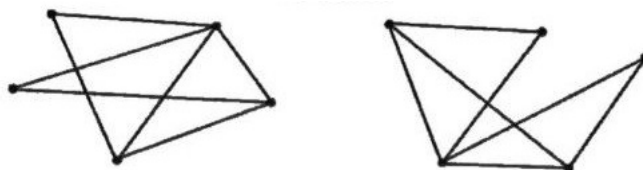
1. Draw the graphs K_6 and $K_{5,3}$.
2. Give an example of a graph with 14 vertices such that (i) Eulerian and Hamiltonian (ii) Eulerian but NOT Hamiltonian (iii) NOT Eulerian but Hamiltonian and (iv) NOT Eulerian and NOT Hamiltonian.
3. Check whether the following two graphs are isomorphic or not.



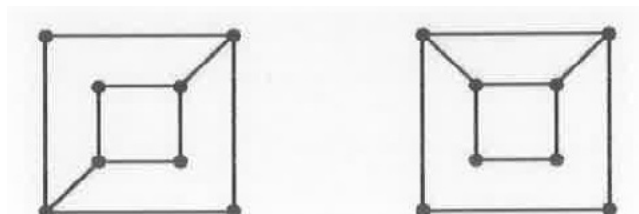
4. Find an Euler path or an Euler circuit, if it exists in each of the following three graphs. If it does not exist, explain why?



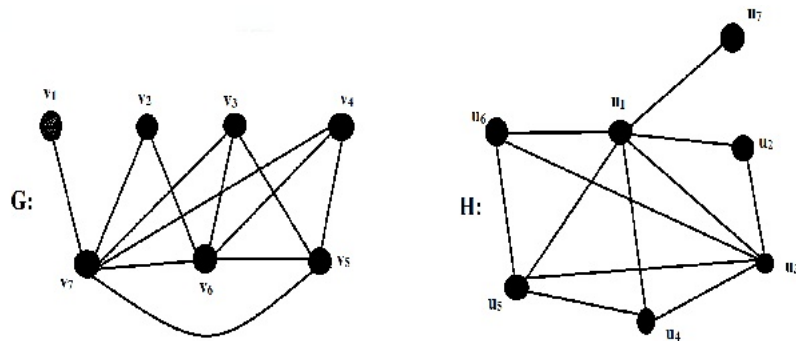
5. Check whether the following two graphs are isomorphic or not.



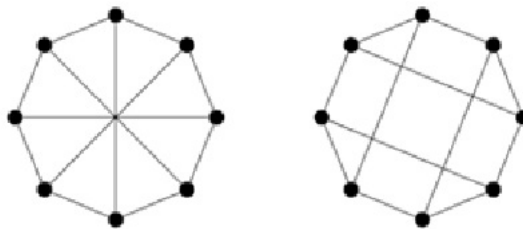
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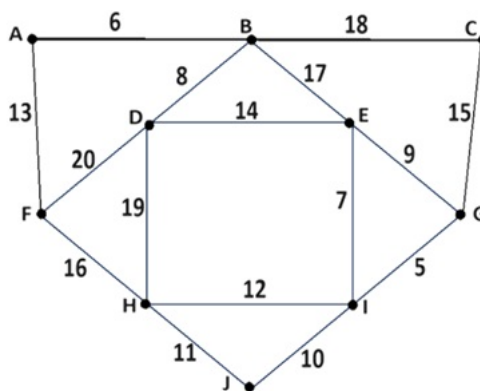
7. Check whether the following two graphs are isomorphic or not.



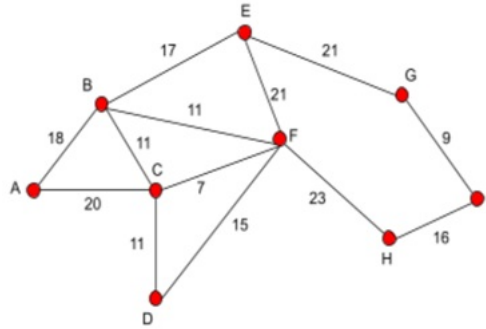
8. Check whether the following two graphs are isomorphic or not.



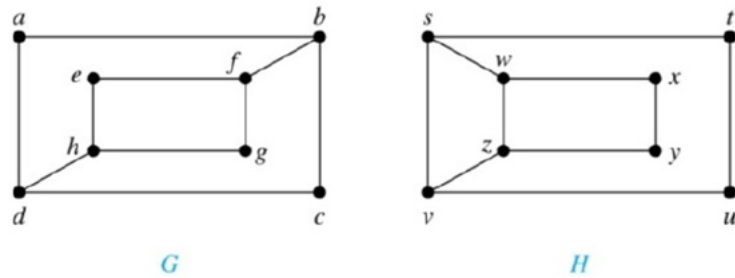
9. A connected, planar graph contains 10 vertices and divides the plane into seven regions. Compute the number of edges in the graph. Give an example of such a graph.
10. Using Kruskal algorithm, find the minimum spanning tree for the following weighted, undirected graph.



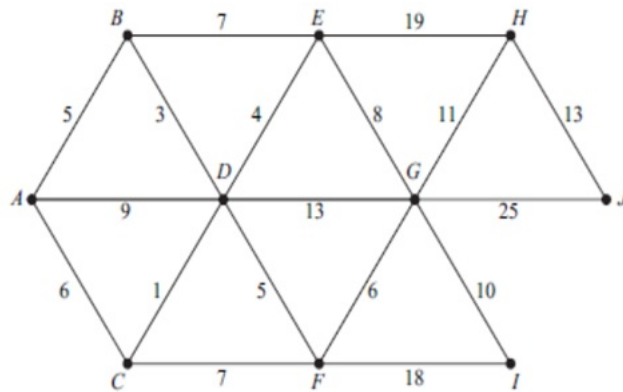
11. Give an example of a planar graph which contains 10 vertices and contains only one planar region.
12. For what values of n , the complete graph K_n be planar? Justify.
13. Find the minimum spanning tree for the following weighted, undirected graph.



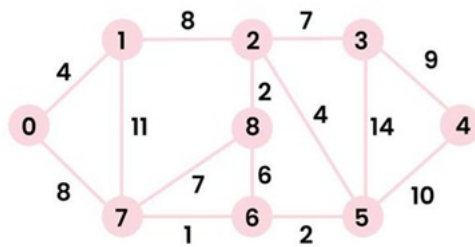
14. Check whether the following two graphs are isomorphic or not.



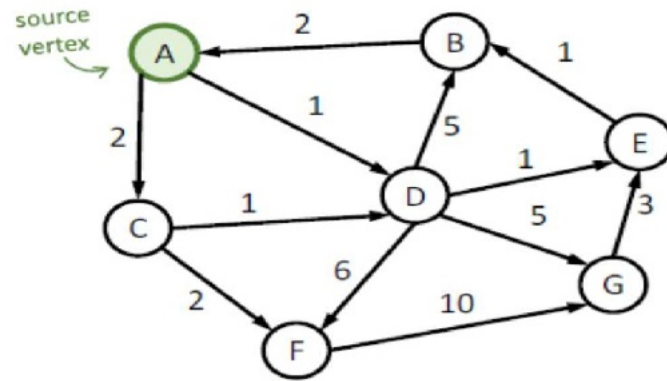
15. Using Kruskal algorithm, find the minimum spanning tree for the following weighted, undirected graph.



16. Using Dijkstra's algorithm, find the shortest distance from the source node 0 to all other nodes the following weighted undirected graph.



17. Using Dijkstra's algorithm, find the shortest distance from the source node A to all other nodes the following weighted directed graph.



18. Define Cut Sets and Fundamental Circuits of a graph. For the following graph, find the cut set and fundamental circuits.

